EFY TESTED
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GETTING TO KNOW IC 555 Through Experiments

Part 2

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The first six circuits to learn the theory and digital logic behind twelve simple experiments that can be performed using IC 555 were published in December 2022 issue. Here are the remaining six circuits.

Although the circuits are non-conventional, they help understand the IC 555 comprising voltage divider, trigger comparator, threshold comparator, SR flip-flop, discharge transistor, reset transistor, and output stage. They also help understand the IC's three modes of operation: monostable, astable, and bistable. The visible operation of the IC in all the twelve experiments is shown by LEDs.

All the experiments are simple and can be performed on a breadboard. A power supply of 5V to 9V DC can be used for the circuits (although according to data sheet of IC 555, 4.5V to 15V may be used). In the following circuits two 555 ICs are used to create interesting situations and make the learning an enjoyable experience.

Experiment 7

As shown in Fig. 7, in this circuit both the ICs, IC12 and IC13, are wired as bistable multivibrators. However, the output pin 3 of each IC is connected to the discharge pin 7 of the other IC through an LED and a current limiting resistor. In normal bistable operation, the discharge pin 7 is not used as there is no capacitor in the circuit.

Initially, when the circuit is

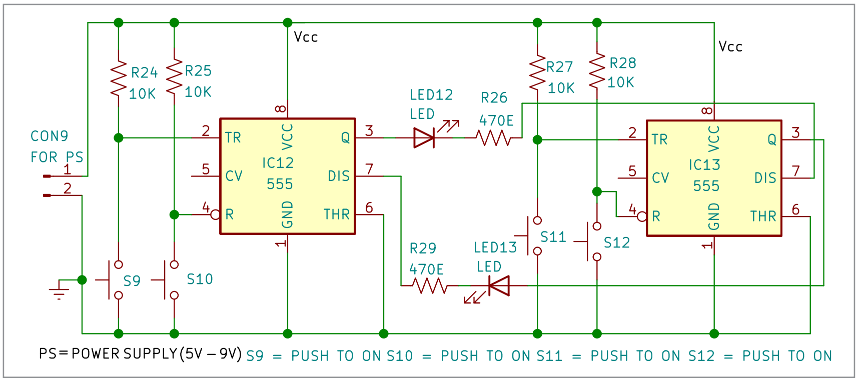

Fig. 7: Circuit diagram for experiment 7

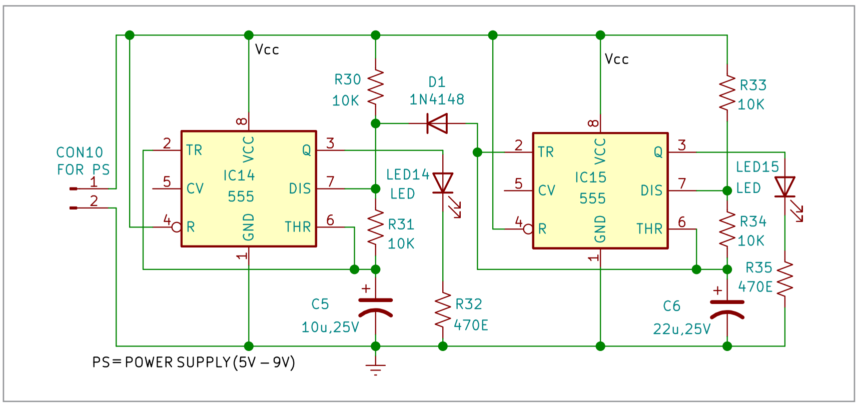

Fig. 8(a): Circuit diagram for first part of experiment 8

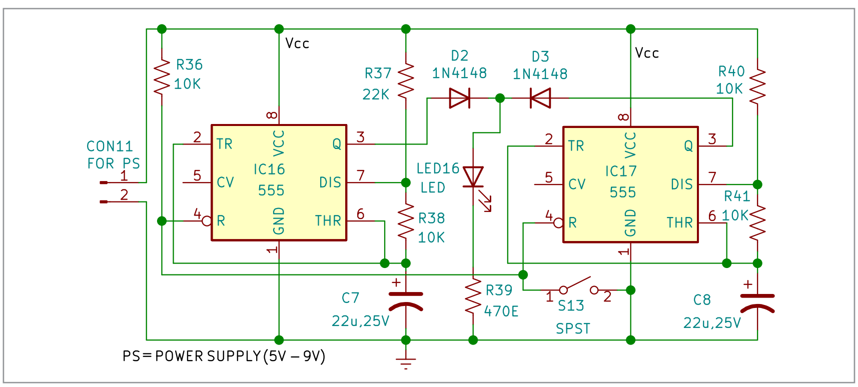

Fig. 8(b): Circuit diagram for second part of experiment 8